

## SIMULATION OF 3-D RANDOM ROUGH SURFACE BY 2-D DIGITAL FILTER AND FOURIER ANALYSIS

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### SUMMARY

*2-D FIR filters are applied to the generation of 3-D random surfaces in this paper. The circularly symmetric low-pass filters are firstly employed to generate the isotropic random surfaces with no restrictions on the shape of ACF. To simulate real rough surfaces, however, one has to generate the surfaces having an expected autocorrelation function and height distribution, which requires determining the filter coefficients corresponding to the specified ACF. A spectrum analysis approach is developed in this paper to compute the filter coefficients and the FFT procedure provides an efficient way to implement the filter. The Johnson translator system of distribution is introduced into the generation procedure of non-Gaussian surface with a required ACF. The analysis indicates that the 2-D FIR filters used in this paper are mathematically identical with the 2-D MA time series model which simulates the ACF within whole correlation regions.*

### 1. INTRODUCTION

The tribological behaviour of engineering surfaces, such as the contact between asperities, hydrodynamic lubrication of rough surfaces and so on, may be predicted numerically. The numerical solution for these problems requires input of surface data, which could be obtained from digital output of profilometer or generated by numerical simulation of the rough surfaces. However, two-dimensional measuring of rough surfaces using a stylus instrument involves special technique and elaborate work while the numerical simulation, if properly designed, is much more efficient and flexible. In addition, the simulation approach provides a convenient means to understand the relation between the geometrical structure of rough surfaces and their statistical description.

The random process description of engineering surfaces has been accepted by Many investigators [1], which makes it possible to generate a rough surface by the random simulation. Patir [2] use "the linear transformation" of a matrix, the components of which are independent random numbers, to generate two-dimensional random surfaces. To determine the transform coefficients, however, requires numerical solution of simultaneous non-linear equations whose convergency is rather questionable. The time series model for random surface characterization [3,4] was applied to one-dimensional profile generation by Watson et al. [5], and later, the Johnson transformation system was introduced into the simulation of non-gaussian distribution with given skewness and kurtosis (Watson and Spedding [6]). The generation of two-dimensional random surfaces was also studied by Whitehouse [7] using AR time series model, but only second order system was considered, which means only the features of autocorrelation function (ACF) near the origin were simulated.

The present study employs 2-D digital filter techniques [8] to deal with the generation of 3-D

random surfaces. The difference equations of the digital filters for random signal is mathematically identical with the expressions of time series model for random process, or more general, it is a linear transformation system of the input signal. Therefore, the present work is an extension of the previous studies, but the techniques in digital filter design such as the spectrum analysis, Fourier transform and the window method make the generation of random surfaces with various specified forms of ACF possible and more efficient.

For the engineering surfaces composed of both random and periodic parts, the two components can be generated separately, and then combined together to produce the simulated surfaces. The present study considers only the generation of random surfaces.

### 2. DIGITAL FILTERS

A digital filter is a system which transforms an input sequence  $\{\eta(I)\}$  to an output sequence  $\{z(I)\}$  [8]. A very important class of linear systems is the one described by the following equation

$$\sum_{k=0}^{n-1} a(k) \eta(I-k) = \sum_{k=0}^{m-1} b(k) z(I-k) \quad (1)$$

where  $a(k)$  and  $b(k)$  are the coefficient which define the system. If the coefficients satisfy the following conditions

$$\begin{aligned} b(0) &= 1 \\ b(k) &= 0, \quad k \neq 0 \end{aligned}$$

the difference equation (1) reduces to

$$z(I) = \sum_{k=0}^{n-1} a(k) \eta(I-k) \quad (2)$$

These filters, which have finite responses of length  $n$  to the unit sample sequence  $\delta(I)$ , are known as finite impulse response filters (FIR) while the filters described by equation (1) are

known as *infinite impulse response filters* (IIR). In the present paper, only FIR filter are considered.

For two-dimensional (2-D) systems, FIR filters are defined as

$$z(I, J) = \sum_{k=0}^{n-1} \sum_{l=0}^{m-1} h(k, l) \eta(I-k, J-l) \quad (3)$$

If we take the Fourier transform of the two sides of eq. (2), we obtain

$$Z(\omega) = H(\omega) A(\omega) \quad (4)$$

where  $A(\omega)$ ,  $Z(\omega)$  are Fourier transforms of the input sequence  $\{\eta(I)\}$  and output sequence  $\{z(I)\}$  respectively, and  $H(\omega)$  is called the *frequency response* or *transfer function* of the system, expressed as

$$H(\omega) = \sum_{k=0}^{n-1} a(k) e^{-jk\omega} \quad (5)$$

Similarly, the Fourier transform of the 2-D linear system (3) gives

$$Z(\omega_x, \omega_y) = H(\omega_x, \omega_y) A(\omega_x, \omega_y) \quad (6)$$

with transfer function

$$H(\omega_x, \omega_y) = \sum_{k=0}^{n-1} \sum_{l=0}^{m-1} h(k, l) e^{-jk\omega_x} e^{-jl\omega_y} \quad (7)$$

Design of a digital filter means determining filter coefficients  $a(k)$  or  $h(k, l)$  to provide the system an expected frequency response. For the most commonly used form of filter such as low-pass, high-pass and so on, the analytical derivation of 1-D coefficients  $a(k)$  is simple, while the 2-D coefficients  $h(k, l)$  may be obtained by certain transformation technique from 1-D filters to 2-D filters. The evaluation of coefficients for the general form of digital filters can be achieved by computing the inverse Fourier transform of a given transfer function  $H(\omega)$  or  $H(\omega_x, \omega_y)$ .

One problem is that the Fourier series so obtained has an infinite number of coefficients, the truncation of the Fourier series in some way is therefore necessary. This truncation unfortunately leads to the Gibbs phenomenon which causes the error in the approximate frequency response near a discontinuity. To minimize the effects of the Gibbs phenomenon, various data windows have been proposed and used for the design of FIR digital filters.

With the filter coefficients obtained, the filtering operation can be implemented on a general purpose digital computer either by the direct convolution expressed in eq. (2) and (3), or by transferring the convolution operation into the frequency domain, using the convolution theorem of Fourier transform as shown in eq. (4) and (6). In most cases, the FFT algorithms make the filtering in the frequency domain much faster than the direct convolution.

When the input of the system is a random sequence, equations (1), (2) and (3) are identical with the expressions of the time series model. In these cases, the IIR filters expressed as eq. (1) corresponds to autoregressive moving average (ARMA) processes and the FIR filters expressed as eq. (2) or (3) correspond to moving average (MA) processes. It is therefore theoretically verified to use digital filter techniques in the simulation of random surfaces.

### 3. GENERATION OF RANDOM SURFACES

Rough surfaces, as random process, are characterized mainly by two statistical functions: the height distribution and the

autocorrelation function (ACF). The discrete samples of the surface amplitude correspond to a 2-D random sequence. To simulate real rough surfaces one has to generate the random sequences having specified form of height distribution and ACF.

Using random number generators, one first generates the input sequences  $\{\eta(I, J)\}$  composed of independent random numbers. The output sequences  $\{z(I, J)\}$  corresponding to the random surfaces are then obtained through the filtering operation (eq. (3)), which gives the sequences a certain form of the ACF. If the input sequences are Gaussian distributed, the distribution will remain unchanged when the sequences pass through the linear system.

For generating non-Gaussian surfaces, it is necessary to specify the skewness and kurtosis of  $\{\eta(I, J)\}$ , which can be estimated according to the relations between the distributions of input and output sequences [6].

#### 3.1 Generation of Isotropic Gaussian Surfaces Using Circularly Symmetric Low-pass Filters

If there is no restriction on the exact shape of the ACF, a simple way for generating the random surfaces is putting the independent random sequence into a low-pass filter in order to modify the correlation aspect of the sequence and provide the generated surface a spatial arrangement.

For the generation of isotropic surfaces, we employ the 2-D circularly symmetric low-pass filters [8], the frequency response of which has the form

$$H(\omega_x, \omega_y) = \begin{cases} 1 & (\omega_x^2 + \omega_y^2)^{1/2} \leq \omega_c \\ 0 & (\omega_x^2 + \omega_y^2)^{1/2} > \omega_c \end{cases} \quad (8)$$

the coefficients of the Fourier series of this function have the form

$$h(k, l) = \alpha \frac{J_1(2\pi\alpha\sqrt{k^2+l^2})}{(k^2+l^2)^{1/2}} \quad (9)$$

where  $\omega_c$  = the cut-off circle frequency  
 $\alpha = f_c/f_s$ , the normalized frequency  
 $f_c = 2\pi\omega_c$ , the cut-off frequency  
 $f_s$  = the sampling frequency  
 $J_1$  = the first order Bessel function

If the input sequence  $\{\eta(I, J)\}$  composed of independent Gaussian random numbers has been generated, the height coordinates of the simulated surface are obtained through the filtering operation

$$z(I, J) = \sum_{k=-n/2+1}^{n/2-1} \sum_{l=-m/2+1}^{m/2-1} h(k, l) \eta(I+k, J+l) \quad (10)$$

$I = 0, 1, \dots, N-1, J = 0, 1, \dots, M-1$

When a window function  $w(k, l)$  is designed to limit the Gibbs phenomenon, the filter coefficients in eq. (10) should be  $h_w(k, l)$  instead of  $h(k, l)$ , where

$$h_w(k, l) = h(k, l)w(k, l) \quad (11)$$

It is a general experience that if  $w(t)$  is a good window in 1-D case, then the 2-D circularly symmetric window obtained from a rotation of  $w(t)$  in the form

$$w(k, l) = w([k^2+l^2]^{1/2}) \quad (12)$$

is also a good 2-D window.

Figure 1 shows the filter coefficient  $h_w(k, l)$  and its frequency response  $H(\omega_x, \omega_y)$  obtained using the Lanczos window with the parameter  $m_w = 1.6$  and  $\alpha = 0.20$ .

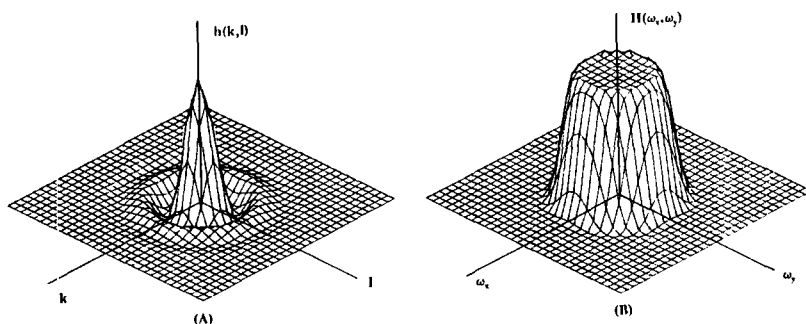


Fig.1 (A) Filter coefficient  $h(k, l)$  and (B) frequency response  $H(\omega_x, \omega_y)$  of a circular low-pass digital filter for  $\alpha = 0.20$ , using Lanczos window  $m_0 = 1.6$  ( $n=m=32$ ).

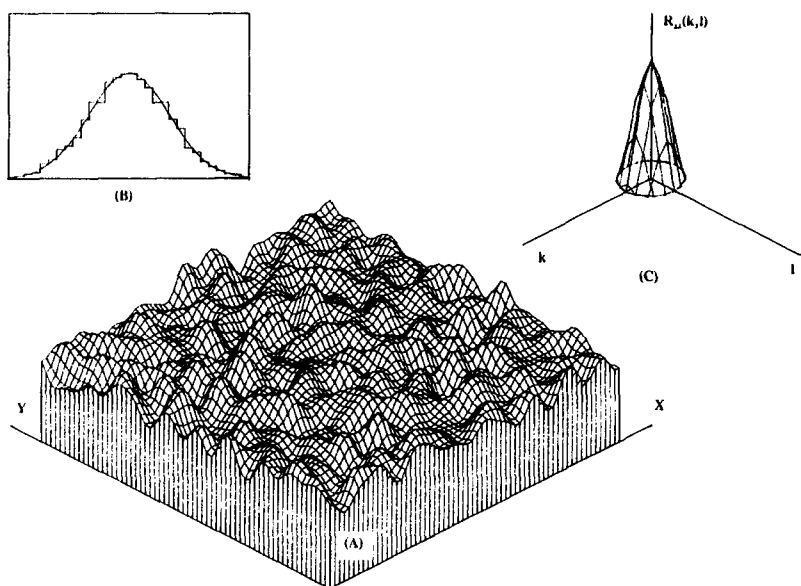


Fig.2 Perspective display of random surface generated using the circular low-pass filter for  $\alpha = 0.20$ . (A) - view of the surface; (B) - height distribution of the generated surface and a standard normal distribution curve; (C) - autocorrelation function of the generated surface.

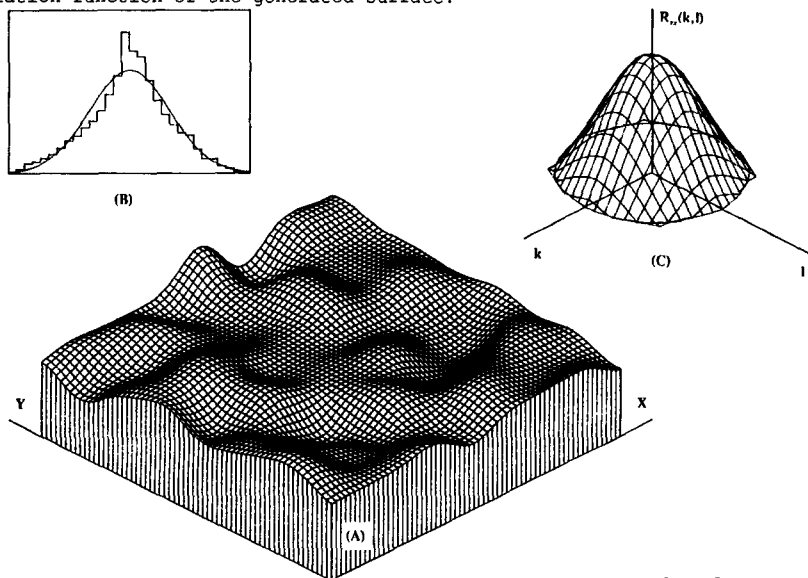


Fig.3 Perspective display of random surface generated using the circular low-pass filter for  $\alpha = 0.05$ . (A) - view of the surface; (B) - height distribution of the generated surface and a standard normal distribution curve; (C) - autocorrelation function of the generated surface.

Figure 2 shows the perspective view of a random surface generated using the circular symmetric low-pass digital filters with  $\alpha = 0.20$ . The height distribution and ACF of the generated surfaces are also displayed on the upper part of the figures. The spatial feature of the generated surfaces may be modified by changing the filtering frequency  $\alpha$ . An example of  $\alpha = 0.05$  is given in Fig.3.

The amplitudes of the generated surfaces have been normalized to satisfy that the r.m.s. roughness  $\sigma = 1$ . To obtain surfaces with a given r.m.s. value, it requires only to multiply the roughness amplitudes by the desired  $\sigma$  value. The other parameters are chosen to be

$$N = M = 64 \quad n = m = 32$$

This arrangement for the values of parameters  $\sigma, N, M, n$  and  $m$  will remain unchanged through the present paper if there is no particular specification.

The results shown in Fig.2 and Fig.3 indicate that it is possible to generate random surfaces by means of digital filter techniques, and the generated surfaces show an isotropic pattern if the circularly symmetric filters are employed.

### 3.2 Generation of Gaussian Surfaces Having Specified Autocorrelation Function

Our purpose is to generate the random sequences which have a specified form of ACF and height distribution to simulate the real rough surfaces.

In the discrete form, the autocorrelation function of stationary random process  $z(I, J)$  is defined to be

$$R_{zz}(k, l) = E\{z(I, J) z(I+k, J+l)\} \quad (13)$$

If the process is an ergodic one, the above expression is identical with

$$R_{zz}(k, l) = \frac{1}{NM} \sum_{I=0}^{N-1} \sum_{J=0}^{M-1} z(I, J) z(I+k, J+l)$$

$$\text{as } N, M \rightarrow \infty. \quad (14)$$

The Fourier transform of  $R$  is called the power spectral density (PSD)

$$S_{zz}(\omega_x, \omega_y) = \frac{1}{nm} \sum_{k=-n/2+1}^{n/2-1} \sum_{l=-m/2+1}^{m/2-1} R_{zz}(k, l) e^{-jk\omega_x} e^{-jl\omega_y} \quad (15)$$

$$\omega_x = -n/2+1, \dots, -1, 0, 1, \dots, n/2-1$$

$$\omega_y = -m/2+1, \dots, -1, 0, 1, \dots, m/2-1$$

If  $S_{\eta\eta}(\omega_x, \omega_y)$  denotes the PSD of input sequence, the relation between  $S_{\eta\eta}$  and  $S_{zz}$  for a linear system have the form

$$S_{zz}(\omega_x, \omega_y) = |H(\omega_x, \omega_y)|^2 S_{\eta\eta}(\omega_x, \omega_y) \quad (16)$$

where  $H(\omega_x, \omega_y)$  is the frequency response or transfer function of the system.

Since  $\{\eta(I, J)\}$  is a sequence composed of independent random number, its PSD must be a constant

$$S_{\eta\eta}(\omega_x, \omega_y) = C \quad (17)$$

For the generation of random surfaces, it is reasonable to design a filter with the frequency response being a real function and the filter coefficients satisfying

$$h(k, l) = h(-k, l) = h(-k, -l) = h(k, -l) \quad (18)$$

According to this discussion, the eq.(16) can be rewritten as

$$H(\omega_x, \omega_y) = (S_{zz}(\omega_x, \omega_y) / C)^{1/2} \quad (19)$$

The inverse Fourier transform of  $H(\omega_x, \omega_y)$  gives the filter coefficients

$$h(k, l) = \frac{1}{nm} \sum_{\omega_x=-n/2+1}^{n/2-1} \sum_{\omega_y=-m/2+1}^{m/2-1} H(\omega_x, \omega_y) e^{-jk\omega_x} e^{-jl\omega_y}$$

$$k = -n/2+1, \dots, -1, 0, 1, \dots, n/2-1$$

$$l = -m/2+1, \dots, -1, 0, 1, \dots, m/2-1 \quad (20)$$

Therefore, the simulation of random surfaces is achieved by following procedures.

(1) Specify the ACF of the simulated surfaces according to assuming expressions or the measurement of engineering surfaces.

(2) Generate 2-D input sequences  $\{\eta(I, J)\}$  with  $I = -n/2+1, \dots, 0, \dots, N-1+(n/2-1)$ ,  $J = -m/2+1, \dots, 0, \dots, M-1+(m/2-1)$ . The components of  $\{\eta(I, J)\}$  are independent Gaussian distributed random numbers.

(3) Compute the Fourier transform of  $R_{zz}(k, l)$  obtaining the power spectral density  $S_{zz}(\omega_x, \omega_y)$ .

(4) Determine the frequency response  $H(\omega_x, \omega_y)$  according to eq.(19).

(5) Compute the inverse Fourier transform of  $H(\omega_x, \omega_y)$  obtaining the filter coefficient  $h(k, l)$ .

(6) Compute the convolution of sequences  $\{\eta(I, J)\}$  and  $\{h(k, l)\}$  using eq.(10), obtaining the 2-D output sequence  $\{z(I, J)\}$  which corresponds to the height coordinates of the simulated surfaces.

No window function is employed in this procedure since it will affect the accuracy of the shape of ACF.

As an example of this procedure, a random Gaussian surface having an exponential ACF is generated.

Assume that the ACF for the simulated surface has the form

$$R_{zz}(k, l) = \sigma^2 \exp(-2.3[(k/\beta_x)^2 + (l/\beta_y)^2]^{1/2}) \quad (21)$$

where  $\sigma$  is the RMS of the surface roughness, and the correlation lengths  $\beta_x, \beta_y$  are defined as the lengths at which the ACFs of the  $x$  and  $y$  profiles reduce to 10% of their values at the origin.

It is convenient for a 2-D ACF to introduce a new term - the correlation regions defined as an area in which the ACF of the surfaces reduces from  $\sigma^2$  at the origin to  $0.1\sigma^2$ . For the ACF shown in eq.(21), the correlation regions is an ellipse. The ratio of the lengths of two axes for the ellipse

$$\gamma = \beta_x / \beta_y \quad (22)$$

is called as the pattern parameter since it shows the degree of non-isotropy of rough surface.

We choose  $\beta_x = 4$ ,  $\beta_y = 12$ , i.e.  $\gamma = 1/3$ , for the present case to generate a non-isotropic Gaussian surface.

The perspective view of the generated surface is displayed in Fig.4. The height distribution and ACF computed according to the output sequence  $\{z(I, J)\}$  are also shown on the upper part of the same figure. The similarity of the ACF of the generated surface to the expected ACF indicates that the procedure presented in this section is able to generate the random surfaces having a specified form of ACF within the correlation regions. The non-isotropic pattern of the generated surface can be clearly observed from a contour map of the surface, as shown in Fig.5.

### 3.3 Generation of Random Surfaces Using FFT

The filtering operation by the direct convolution, as shown in eq.(10), is simple but not very efficient from the point of view of

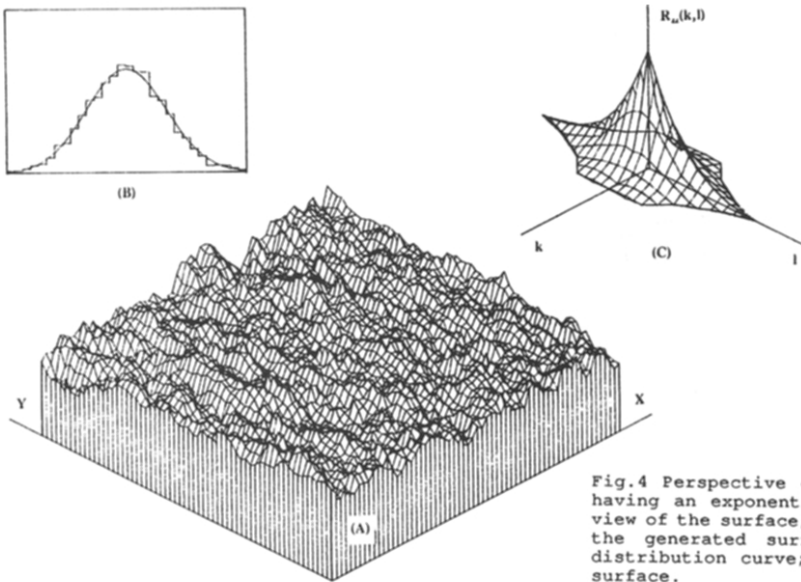


Fig.4 Perspective display of generated surface having an exponential ACF with  $\gamma = 1/3$ . (A) - view of the surface; (B) - height distribution of the generated surface and a standard normal distribution curve; (C) - ACF of the generated surface.

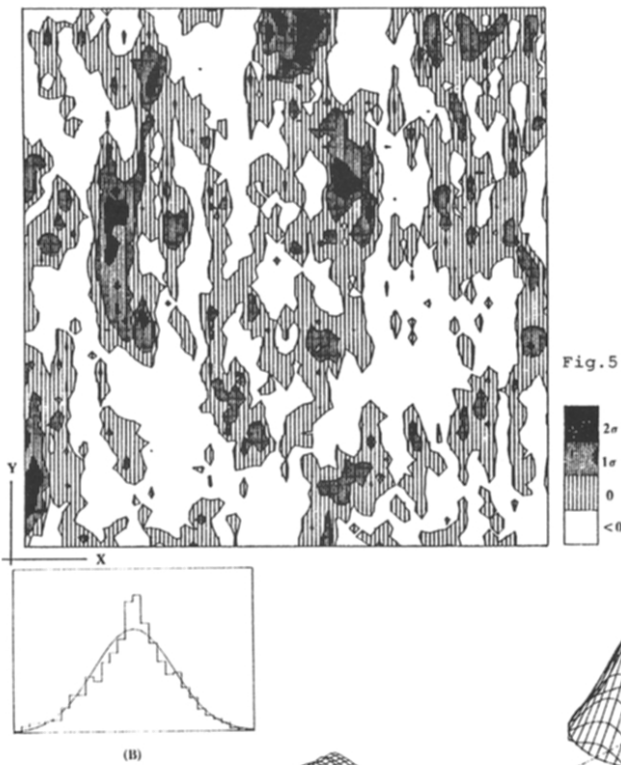


Fig.5 Contour map of the generated surface in Fig.4

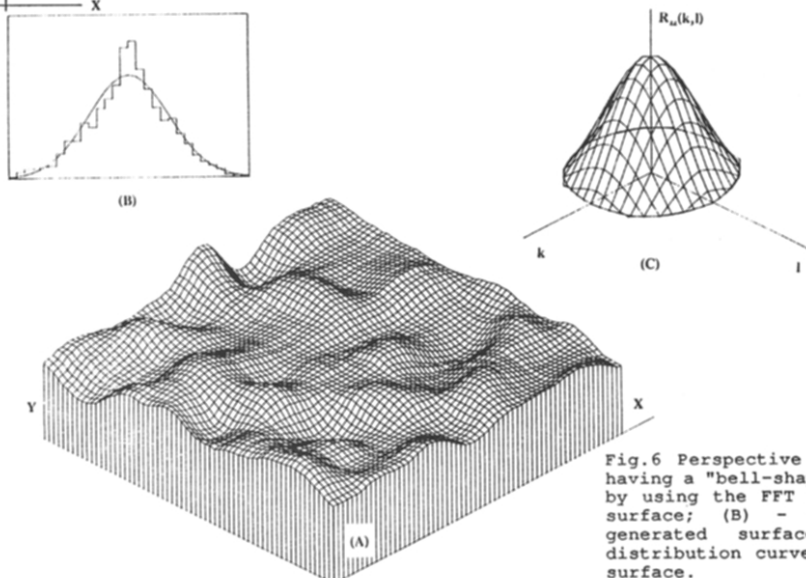


Fig.6 Perspective display of a random surface having a "bell-shaped" ACF with  $\gamma = 1$ , generated by using the FFT procedure. (A) - view of the surface; (B) - height distribution of the generated surface and a standard normal distribution curve; (C) - ACF of the generated surface.

processing speed. A second and more efficient method to perform FIR filtering operations is based on the frequency domain approach and FFT algorithms.

Assume that we have an input sequence  $\{\eta(I, J)\}$ ,  $I = 0, \dots, N-1$ ,  $J = 0, \dots, M-1$ , which has to be filtered by means of a FIR filter to generate an output sequence  $\{z(I, J)\}$  whose ACF has been specified as  $R_{zz}(k, l)$ ,  $k = 0, \dots, n/2-1$ ,  $l = 0, \dots, m/2-1$ .

If  $L_x$  and  $L_y$  are the minimum powers of 2 greater than  $N+n-1$ ,  $M+m-1$  respectively, it is possible to perform the generating in the frequency domain using the following procedure.

(1) construct a new sequence  $\{\eta_t(I, J)\}$ ,  $I = 0, \dots, L_x-1$ ,  $J = 0, \dots, L_y-1$ , where

$$\{\eta_t(I, J)\} = \{\eta(I, J)\}, \quad I = 0, \dots, N-1, J = 0, \dots, M-1$$

$$\{\eta_t(I, J)\} = 0 \quad I = N, \dots, L_x-1, J = M, \dots, L_y-1$$

(23)

(2) construct a new ACF sequence  $\{R_t(I, J)\}$ ,  $I = 0, \dots, L_x-1$ ,  $J = 0, \dots, L_y-1$ , where

$$\begin{cases} \{R_t(I, J)\} = \{R_{zz}(I, J)\} \\ I = 0, \dots, n/2-1, J = 0, \dots, m/2-1 \\ \{R_t(L_x-I+1, J)\} = \{R_{zz}(I, J)\} \\ \{R_t(I, L_y-J+1)\} = \{R_{zz}(I, J)\} \\ \{R_t(L_x-I+1, L_y-J+1)\} = \{R_{zz}(I, J)\} \\ I = 1, \dots, n/2-1, J = 1, \dots, m/2-1 \\ \{R_t(I, J)\} = 0 \quad \text{otherwise} \end{cases} \quad (24)$$

(3) Compute the 2-D FFT of  $\{\eta_t(I, J)\}$  obtaining the sequence  $\{A_t(\omega_x, \omega_y)\}$ ,  $\omega_x = 0, \dots, L_x-1$ ,  $\omega_y = 0, \dots, L_y-1$

(4) Compute the 2-D FFT of  $\{R_t(I, J)\}$  obtaining a real sequence  $\{S_{zz}(\omega_x, \omega_y)\}$ ,  $\omega_x = 0, \dots, L_x-1$ ,  $\omega_y = 0, \dots, L_y-1$

(5) Determine the frequency response of the filter  $\{H_t(\omega_x, \omega_y)\}$  using eq.(19),  $\omega_x = 0, \dots, L_x-1$ ,  $\omega_y = 0, \dots, L_y-1$

(6) Compute the complex product of two sequences  $\{H_t(\omega_x, \omega_y)\}$  and  $\{A_t(\omega_x, \omega_y)\}$ , obtaining a new sequence  $\{Z(\omega_x, \omega_y)\}$ ,  $\omega_x = 0, \dots, L_x-1$ ,  $\omega_y = 0, \dots, L_y-1$

(7) Compute the 2-D IFFT of the sequence  $\{Z(\omega_x, \omega_y)\}$  obtaining the output sequence  $\{z(I, J)\}$ ,  $I = 0, \dots, N-1$ ,  $J = 0, \dots, M-1$ , which has the specified ACF  $R_{zz}(k, l)$ .

As an example, the FFT procedure has been applied to the generation of a random surface whose ACF has a "bell-shaped" form

$$R_{zz}(k, l) = \sigma^2 \exp\{-2.3[(k/B_x)^2 + (l/B_y)^2]\} \quad (25)$$

where  $\sigma$ ,  $B_x$  and  $B_y$  have the same definitions as in section 3.2. The correlation lengths are chosen to be  $B_x = B_y = 10$ , i.e., the generated surface is isotropic one whose perspective view is shown in Fig.6.

Since  $N=M=64$  and  $n=m=32$  have been fixed in this paper, then we have  $L_x = L_y = 128$ . For this arrangement of parameters, the FFT procedure takes about 30% processing time as the direct convolution. It is possible to increase the values for  $N$  and  $M$  up to 95 without wraparound error, the efficiency of the FFT procedure will be more impressive in that case.

Fig.6 shows a similar surface in comparison with Fig.3. The reason is not only that both surfaces have isotropic pattern and a similar ACF, but also that we employ the same input sequence  $\{\eta(I, J)\}$  for the two procedures. To obtain different surfaces with similar statistical properties, one should simply change the seeds in generating the input sequences.

Moreover, the similarity between Fig. 6 and Fig.3 provides an evidence that the FFT procedure will produce the same results as the direct convolution.

As a comparison, Figure 7 shows another random surface generated by the same procedure but with an ACF in cone form

$$R_{zz}(k, l) = \sigma^2 \{1 - [(k/B_x)^2 + (l/B_y)^2]^{1/2}\} \quad (26)$$

The surfaces in Fig.6 and Fig.7 display quite different appearance despite their similarity in the form and size of the correlation regions.

### 3.4 Generation of Non-Gaussian Random Surfaces

For the generation of non-Gaussian random surfaces, the distribution of output sequences will not be of the same form as that of input sequences in most cases. Since there is no general relationship available between input and output distributions, one practical solution is to simulate the skewness and kurtosis of the distribution using the procedure presented by Watson and Spedding [6].

For a pure MA process

$$z_t = \theta_0 \eta_t + \theta_1 \eta_{t-1} + \theta_2 \eta_{t-2} + \dots + \theta_q \eta_{t-q} \quad (27)$$

$$(\text{Sk}_z)^{1/2} = \frac{\sum_{i=0}^q \theta_i^3}{(\sum_{i=0}^q \theta_i^2)^{3/2}} (\text{Sk}_\eta)^{1/2} \quad (28)$$

$$\text{Ku}_z = \frac{\sum_{i=0}^q \theta_i^4 \text{Ku}_\eta + 6 \sum_{i=0}^{q-1} \sum_{j=i+1}^q \theta_i^2 \theta_j^2}{(\sum_{i=0}^q \theta_i^2)^2} \quad (29)$$

where  $\text{Sk}_z$ ,  $\text{Sk}_\eta$ ,  $\text{Ku}_z$  and  $\text{Ku}_\eta$  are the skewness and kurtosis of the distributions for random processes  $z_t$  and  $\eta_t$ .

The difference equation (3) for 2-D FIR filters may be rewritten as the same form as eq.(27) by introducing

$$\begin{cases} \eta_t = \eta(I, J) \\ z_t = z(I, J), \quad I=0, \dots, N-1, J=0, \dots, M-1, t=IM+J \\ \theta_i = h(k, l), \quad k=0, \dots, n-1, l=0, \dots, m-1, i=km+l \end{cases} \quad (30)$$

If the distribution of the surfaces to be simulated have been specified, the skewness and kurtosis of the input sequence can be determined according to eq.(28), (29) and (30). Hence the procedure for non-Gaussian surfaces generation is as follows

(1) Determine the skewness  $\text{Sk}_z$  and kurtosis  $\text{Ku}_z$  of output sequences according to the specified distribution.

(2) Compute the skewness  $\text{Sk}_\eta$  and kurtosis  $\text{Ku}_\eta$  of input sequence using eq.(28) and (29).

(3) Generate a Gaussian distributed sequence using a random normal generator.

(4) Transform the normal sequence to the input sequence  $\{\eta(I, J)\}$ , which has a distribution with appropriate skewness  $\text{Sk}_\eta$  and kurtosis  $\text{Ku}_\eta$ , by using the Johnson translator system of distributions [10]. The Johnson distribution curves are fitted according to the  $\text{Sk}_\eta$  and  $\text{Ku}_\eta$  using the algorithm given by Hill et al [9].

(5) Determine the filter coefficient  $h(k, l)$  by using the method presented in section 3.2.

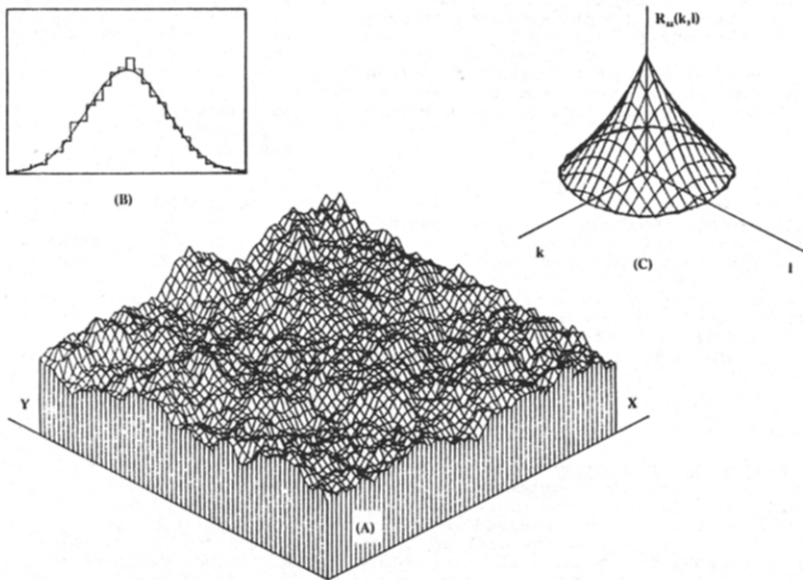


Fig.7 Perspective display of a random surface having a "cone-shaped" ACF with  $\gamma = 1$ , generated by using the FFT procedure. (A) - view of the surface; (B) - height distribution of the generated surface and a standard normal distribution curve; (C) - ACF of the generated surface.

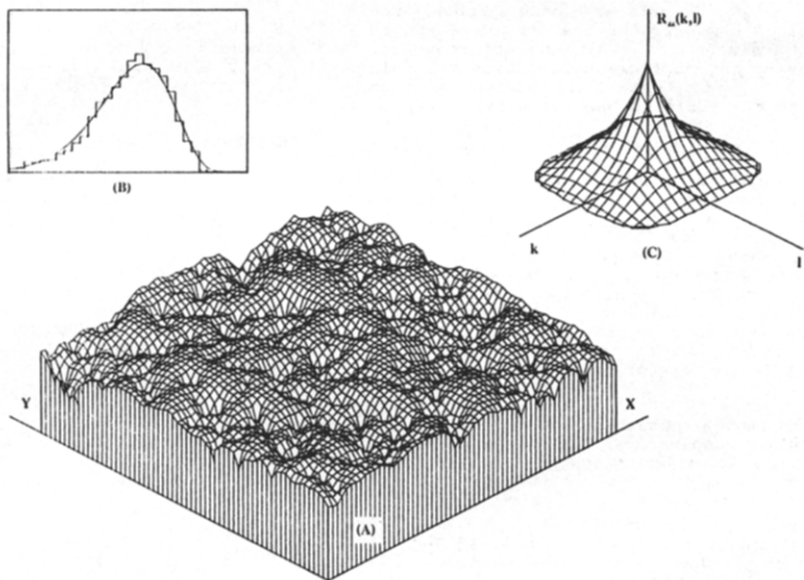


Fig.8 Perspective display of a random surface having a Gamma distribution and an exponential form of ACF with  $\gamma = 1$ ,. (A) - view of the surface; (B) - height distribution of the generated surface and a theoretical Gamma distribution curve with  $Sk=-0.75$ ; (C) - ACF of the generated surface.

(6) Perform the filtering operation by means of direct convolution or FFT procedure to produce the output sequence  $\{z(I, J)\}$ .

Figure 8 gives an example of non-Gaussian surface generation. The generated surface is specified having the Gamma distribution with the moments

$$\begin{aligned}\sigma &= 1 & Sk_x &= -0.75 \\ Ku_x &= 3 + 1.5Sk_x^2 & &= 3.84375\end{aligned}\quad (31)$$

The specified ACF for the surfaces has an exponential form as show in eq.(21) with the correlation lengths  $\beta_x = \beta_y = 10$ .

The Histogram in Fig.8 (B) shows the actual distribution calculated according to the output sequence  $\{z(I, J)\}$  and the curve gives the theoretical Gamma distribution of  $Sk = -0.75$  for comparison.

#### 4. DISCUSSION

In the present study, the output sequences of the FIR filters are taken as a representation of random surfaces. The questions are, to which extent this simulation is reliable and where are its limitations. As mentioned earlier, the difference equations for the digital filters are mathematically identical with the expressions of the time series model. Some theoretical results of time series analysis are therefore helpful in discussing these questions.

Taking the output sequence of the FIR filter to simulate random surfaces means that the amplitudes of the surfaces are considered to be an moving average of a "white noise" sequence. The necessary and sufficient condition for this consideration is that the autocorrelation function of the surfaces has to satisfy [11]

$$R_{xx}(k, l) = 0, \quad |k| > n/2 \text{ and } |l| > m/2 \quad (32)$$

Two statements can be made from this condition. The first is that the simulating approach presented in this paper is only useful for the random surfaces whose ACFs gradually reduce from  $\sigma^2$  to zero as the delay lengths  $k$  and  $l$  increase, but not for those composed surfaces with a large periodic part. The second, the orders of the filter or MA-model depend on the size of the correlation regions since the condition (32) will not be satisfied unless the area determined by  $n/2$  and  $m/2$  are greater than the correlation regions.

AR-model is another possible time series model in random surfaces generation. For the 2-D AR-model, however, only lower order systems are available because of the difficulty in dealing with the Yule-Walker equations [7].

Assume that  $R_{xx}(k, l)$  denotes the ACF of the surfaces to be simulated and  $R_{xx}(k, l)$  is the ACF of the output sequences of a 2-D AR-model. The analysis indicates that

$$R_{xx}(k, l) = R_{xx}(k, l), \quad 0 \leq k \leq n, \quad 0 \leq l \leq m \quad (33)$$

where  $n, m$  are the orders of the AR-model.

In general, the ACF of the surfaces generated by AR-model is consistent with the surfaces to be simulated only within the regions determined by the orders  $n$  and  $m$ . It is therefore concluded that the MA-model employed in the present study simulates the ACF of random surfaces within the correlation regions while the lower order AR-model used by some other investigators simulates the ACF near the origin.

The features of the ACF at the origin are of great importance for characterizing the rough surfaces, especially for describing the shapes of the asperities. The surfaces shown in Fig.6 and Fig.7 display quite different appearance though both of them have same Gaussian distribution and similar form and size of the correlation regions. This difference is best distinguished by the

shape of the ACF at the origin. However, changes in the form and size of the correlation regions may also affect the main structure and pattern of the random surfaces. For example, the different appearance of the surfaces shown in Fig.2 and Fig.3 result from two sources, one is the differences in the ACF shape at the origin and the other is the different size of the correlation regions.

#### 5. CONCLUSION

The 2-D FIR digital filters are employed in the present work to generate the random surfaces.

(1) If there is no restrictions placed on the shape of the ACF, the isotropic random surfaces can be generated by using the 2-D circularly symmetric low-pass filters. The spatial features of the generated surfaces may be modified by changing the filtering frequency  $\alpha$ .

(2) The method of determining the filter coefficients developed in this paper presents a useful approach to simulate real rough surfaces by means of generating the random surfaces having a specified form of ACF.

(3) It has been shown that the FFT procedure is a more efficient way than the direct convolution in the implementation of filtering operation.

(4) The Johnson translator system of distribution has been introduced into the generation of non-Gaussian random surfaces with both ACF and height distribution specified.

(5) The 2-D FIR filters used in this paper are mathematically identical with the 2-D MA time series model which simulates the ACF within the correlation regions.

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